MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES
(A Constituent unit of MAHE, Manipal)

Cryptology – 6_Theory

Reg. Number	Name	Branch
261100690006	MOHAMMED HARIS	Cybersecurity

MANIPAL SCHOOL OF INFORMATION SCIENCES
MANIPAL
(A constituent unit of MAHE, Manipal)

Q: Cipher – EVIRE.

$P = (C - K) \bmod 26$

All 26 keys possible solution:

K	Plaintext	K	Plaintext
0	EVIRE	13	RIVER
1	DUHQD	14	QHUDQ
2	CIGPC	15	PGICP
3	BSFOB	16	OFSBO
4	ARENA	17	NERAN
5	ZQDMZ	18	MDQZM
6	XOBKX YPCLY	19	LCPYL
7	WNAJW XOBKX	20	KBOXK
8	VMZIV WNAJW	21	JANWJ
9	VMZIV	22	IZMVI
10	ULYHU	23	HYLUH
11	TKXGT	24	GXKTG
12	SJWFS	25	FWJSF

Two worked trials:

K=4

$E = (4-4) \bmod 26 = 0 \rightarrow A$.
$V = (21-4) \bmod 26 = 17 \rightarrow R$
$I = (8-4) \bmod 26 = 4 \rightarrow E$
$R = (17-4) \bmod 26 = 13 \rightarrow N$
$E = (4-4) \bmod 26 = 0 \rightarrow A$

ARENA

K=13

$E = (4-13) \bmod 26 = 17 \rightarrow R$
$V = (21-13) \bmod 26 = 8 \rightarrow I$
$I = (8-13) \bmod 26 = 21 \rightarrow V$
$R = (17-13) \bmod 26 = 4 \rightarrow E$
$E = (4-13) \bmod 26 = 17 \rightarrow R$

RIVER

__/__/____

a) So ARENA & RIVER are the only meaningful plaintexts.

b) · RIVER = TIBER River
ARENA = COLOSSEUM

Both are meaningful, so Antony can't decide from the ciphertext alone.

c) A ciphertext only attack can give more than one meaningful plaintext when the ciphertext is short.

A ~~for~~ longer ciphertext is needed for unique answer.

CPA

Q HAHAHA → NONONO
(CT)

a) H = 7 N = 13
A = 0 O = 14

Affine cipher ⇒ $C = (K_1 \cdot P + K_2) \bmod 26$

~~$K_1 = a$, $K_2 = b$~~ $K_1 = a$ $K_2 = b$

b) $13 = (a \cdot 7 + b) \bmod 26$ — ①
$14 = (a \cdot 0 + b) \bmod 26$ — ②

$-1 = 7a$
$a = 7^{-1}$
$a = (-7^{-1}) \bmod 26$
$a = -15 \bmod 26$

c) $a = 11$

~~$(14 = b) \bmod 26$~~ $14 = b \bmod 26$
~~$b =$~~ $b = 14 \bmod 26$.
$b = 14$

d) $C = 11 \times 7 + 14 \bmod 26$
$C = 13$
$C = 13$.

Q. CRWNZ was encrypted using Affine cipher. mod 26.
~~PT~~. Attacker knows the PT begins with HA.

a)

PT	CT
H = 7	C = 2
A = 0	K = 17

$$C = (a \cdot P + b) \bmod 26$$

$$2 = 7 \times a + b \bmod 26$$
$$17 = 0 \times a + b \bmod 26$$

$$-15 = 7a$$
$$a = \frac{-15}{7}$$
$$a = -15 \times 7^{-1}$$
$$a = -15 \times 15 \bmod 26$$
$$a = 9$$

$$\gcd(9, 26) = 1$$

$$b = 17 \bmod 26$$
$$b = 17$$

b)

$$C = (K_1 P + K_2) \bmod 26$$

$$C = 9 \cdot P + 17 \bmod 26$$

c)

$$P = k^{-1} \cdot (C - K_2) \bmod 26$$

For W (22)

$P = K_1^{-1} \cdot (C - K_2) \bmod 26$

$P = 9^{-1} \cdot (22 - 17) \bmod 26$

$P = 3 \cdot (5) \bmod 26$

$P = 15 \Rightarrow P$

For Z (25)

$P = 9^{-1} (25 - 17) \bmod 26$

$P = 3(8) \bmod 26$

$P = 24 \Rightarrow Y$

Plaintext is HAPPY

6

a) Plaintext RIVER: R=17, I=8, V=21, E=4, R=17
NQXTP key: N=13, Q=16, X=23, T=19, P=15

b) Encrypt

$$C_i = (P_i + K_i) \bmod 26$$

R+N ⇒ $(17+13) \bmod 26 = 4 \rightarrow E$
I+Q ⇒ $(8+16) \bmod 26 = 24 \rightarrow Y$
V+X ⇒ $(21+23) \bmod 26 = 18 \rightarrow S$
E+T ⇒ $(4+19) \bmod 26 = 23 \rightarrow X$
R+P ⇒ $(17+15) \bmod 26 = 6 \rightarrow G$

Ciphertext ⇒ EYSXG

c) Decrypt.

$P_i = (C_i - K_i) \bmod 26$

$(4-13, 24-16, 18-23, 23-19, 6-15) \bmod 26 = (17, 8, 21, 4, 17)$

The value corresponds to RIVER

d) Randomness:

A truly random key makes ciphertext unpredictable; patterns in a non random key ~~#~~ can reveal patterns in the message.

Equal: Each plaintext character needs its own key character. A shorter key must be reused ~~once~~ repeated, which can create exploitable patterns.

Using only once: Reusing a key lets attackers compare ciphertexts and potentially learn relationships b/w the plaintexts.